

Cigarette Use by Sex Among Male and Female Students (YRBS 2007)

1. Research Question

Is the proportion of current cigarette use different between male and female students?

2. Data and Variables



Dataset: YRBS 2007

Sample Size: $n = 13,312$

Group Variable:

WhatsYourSex (Male = 1, Female = 2)

Response Variable:

CurrentCigarette Use (1 = Yes, 0 = No)

3. Statistical Method

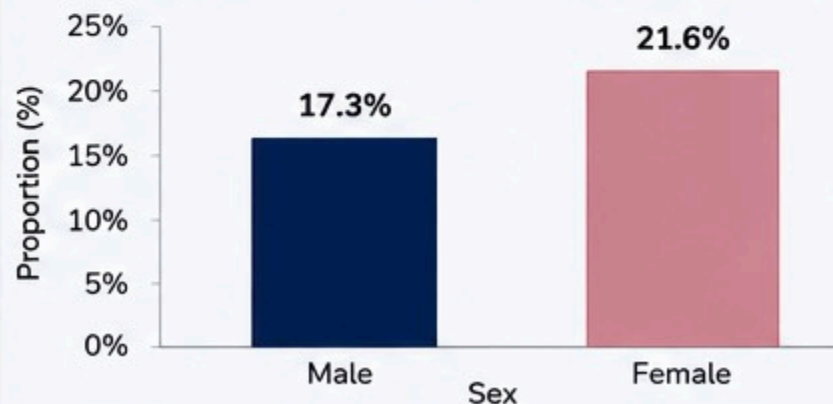


Two-Proportion z-test

We compare the proportions of current cigarette use between male and female students.

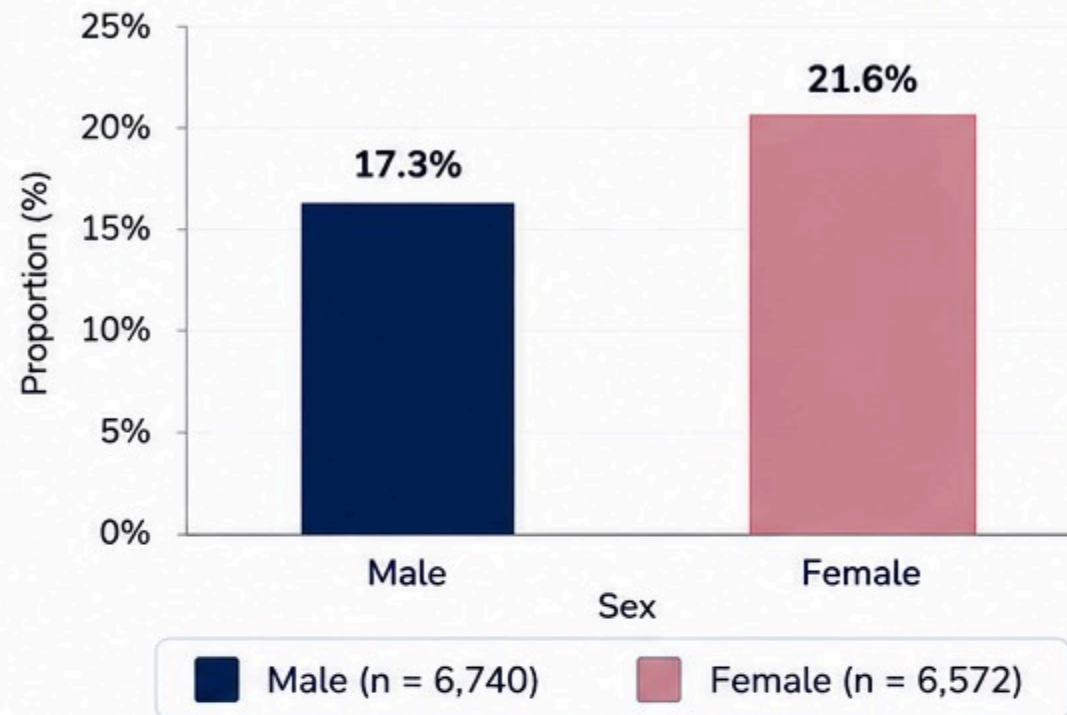
4. Exploratory Data Analysis

Proportion of Current Cigarette Use by Sex



Females have a higher proportion of current cigarette use compared to males.

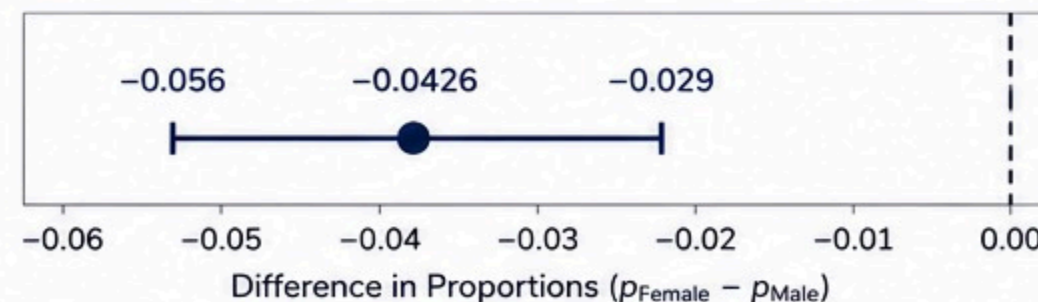
Proportion of Current Cigarette Use by Sex



Sample Summary

Sex	Sample Size (n)	Current Cigarette Use	
		Yes (x)	Proportion ($\hat{p}$)
Male	6,740	1,167	17.3%
Female	6,572	1,418	21.6%
Total	13,312	2,585	19.4%

95% Confidence Interval for $(p_{\text{Female}} - p_{\text{Male}})$



The 95% CI does not include 0, indicating a statistically significant difference.

5. Statistical Test Results

Test	Two-Proportion z-test
Null Hypothesis (H_0)	$p_{\text{male}} = p_{\text{Female}}$
Alternative Hypothesis (H_a)	$p_{\text{male}} \neq p_{\text{Female}}$
z-statistic	-6.215
p-value	5.14×10^{-10}
Significance Level (α)	0.05
Decision	Reject H_0

The p-value is much less than 0.05.
We reject the null hypothesis.

6. Interpretation

- The proportion of current cigarette use among female students (21.6%) is significantly higher than that among male students (17.3%).
- The difference in proportions is **-0.0426**.
- The 95% CI (-0.056, -0.029) suggests that the true difference is likely between -5.6% and -2.9%.

7. Conclusion

Statistical Verdict: At the 5% significance level, we reject the null hypothesis.

There is a **statistically significant difference** in the proportion of current cigarette use between male and female students.

Core Insight: Female students are more likely to use cigarettes than male students in this sample. This finding may reflect gender differences in health behaviors and warrants further investigation.